1 Jarvis and Mckenzie, 1980 General equations

Conservation of mass, momentum and energy:

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$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0 \tag{1}$$

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla P + \rho g + \frac{\partial}{\partial x_j} \tau_{ij} \tag{2}$$

$$\rho C_p \left\{ \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \boldsymbol{u} \cdot [\nabla T - (\nabla T)_s] \right\} = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial u_{ij}} \tag{3}$$

All terms and symbols used in these equations are shown in table 2. The (Newtonian) relation between the deviatoric stress tensor τ_{ij} and the velocity field is

$$\tau_{ij} = \eta \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) + \eta_2 \delta_{ij} \frac{\partial u_k}{\partial x_k}. \tag{4}$$

Combining Eq. (4) with the momentum equation (2) gives the general Navier-Stokes equations, but Jarvis and Mckenzie (1980) does not show these. Instead, they assume that $\frac{1}{Pr} \to 0$, which they assume (do they?) is the case for the Earth's mantle, the inertial forces can be ignored. The resulting equations will be discussed in the next section, which covers the approximations made by Jarvis and Mckenzie (1980).

2 Jarvis and Mckenzie, 1980 Approximations

Problem statement: for the Boussinesq approximation to be valid, the depth range of convection d must be small compared to the temperature scale height H_T ($d/H_T \ll 1$). This is not the case for many systems of deep convection, amongst which the convection of planetary mantles. Physically, $d/H_T \approx O(1)$ means that the influence of viscous dissipation, variable adiabatic temperature- and density gradients must be maintained in the set of governing equations.
The assumptions made for the **anelastic liquid approximation** are that $d = O(H_T)$ and that $\alpha T \ll 1$ what does this mean and how does this relate to Prandtl number? and if $\alpha \ll 1$, how come D is not $\ll 1$?

2.1 Reference state and equation of state

A reference state with hydrostatic pressure P_H and adiabatic temperature distribution (gradient?) T_s is considered. The equation of state is written as a linear Taylor expansion of ρ about the reference state density distribution ρ_r:

$$\rho(T, P) = \rho_r [1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \tag{5}$$

Where

$$\alpha = \frac{1}{\rho_r} \left(\frac{\partial \rho_R}{\partial T} \right)_P \tag{6}$$

$$K_T = \rho_r \left(\frac{\partial P}{\partial \rho_r} \right)_T \tag{7}$$

Determining the reference density distribution ρ_r is done by considering the gradient of $\rho_r(S, P_H)$ with S the entropy per unit mass:

$$\nabla \rho_R(S, P_H) = \left(\frac{\partial \rho_r}{\partial S} \right)_{P_H} \nabla S + \left(\frac{\partial \rho_r}{\partial P_H} \right)_S \nabla P_H \tag{8}$$

With $\nabla S = 0$ since the reference state is adiabatic, and with $\nabla P_H = -\rho_r g \hat{\boldsymbol{z}}$:

$$\frac{1}{\rho_r} \frac{\partial \rho_r}{\partial z} = -\frac{\rho_r g}{K_s} \tag{9}$$

with $K_s = \rho_r \left(\frac{\partial P_H}{\partial \rho_r}\right)_s$. Using the definition of the Grüneisen parameter Γ and the temperature scale height of the liquid H_T gives the Adams-Williamson relation add images!

$$\frac{1}{\rho_r}\frac{\partial \rho_r}{\partial z} = -\frac{1}{H_T \Gamma} \tag{10}$$

Which is valid for a **chemically homogeneous adiabatic density distribution under hydrostatic pressure**. Integration of equation 10 over z gives the reference densitiy distribution as

$$\rho_r(z) = \rho_0 \exp[(d-z)/H_T\Gamma] \tag{11}$$

where $\rho_0 = \rho_r(z = d)$ is the reference density value at the surface orientation of coordinate system?? Plugging this expression of the reference density distribution into 5 gives the (linearized equation) of state

$$\rho(T, P) = \rho_0 \exp[(d-z)/H_T\Gamma][1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \tag{12}$$

2.2 Nondimensionalization

3, 3. move tables here? In this section, Jarvis and Mckenzie, 1980 introduces various characteristic values of variables in order to nondimensionalize the preceding equations. These characteristic values and the resulting dimensionless variables are presented in the tables below.The characteristic value for **velocity** is taken directly from McKenzie et al., 1974, and their derivation is expanded below:

Our starting point is the momentum equation (in the notation of McKenzie et al., 1974), with the left-hand side set to 0 (on the basis of $\mathrm{Re} \ll 1$ and $\frac{1}{\mathrm{Pr}} \to 0$):

$$0 = -\frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\partial \phi}{\partial x_i} + \frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} \tag{13}$$

with

$$\tau_{ij} = \rho\nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$
$$\nabla\phi = \boldsymbol{g}$$

last term in Eq. (13) becomes help index confusion

$$\begin{aligned}\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} &= \rho\nu\left(\frac{\partial}{\partial x_j}\frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j}\frac{\partial u_j}{\partial x_i}\right) \\ &= \rho\nu(\nabla^2 u_i + \frac{\partial}{\partial x_i}\nabla\cdot\boldsymbol{u})\end{aligned}$$

assuming incompressibility, $\nabla \cdot \boldsymbol{u} = 0$, and or should this be $\boldsymbol{u}$ (vector)?

$$\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = \nu\nabla^2 u_i \tag{14}$$

Now we formulate an equation of state under the Boussinesq approximation (Taylor expansion of density $\rho(T)$ around a reference state temperature T_0):

$$\begin{aligned}\rho &= \rho_0[1 - \alpha(T - T_0)] \\ \frac{1}{\rho} &= \rho_0^{-1}[1 - \alpha(T - T_0)]^{-1} \\ &\approx \rho_0^{-1}[1 + \alpha(T - T_0)]\end{aligned}$$

with $\alpha = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_P$ the thermal expansivity coefficient.

We can split the pressure gradient ∇P into a hydrostatic part (∇P_s) and a non-hydrostatic part ∇p:

$$\begin{aligned}\nabla P &= \nabla P_s + \nabla p \\ &= \rho_0 \boldsymbol{g} + \nabla p\end{aligned}$$

Together with the equation of state, the pressure term + gravity in Eq. (13) then becomes

$$\begin{aligned}-\frac{1}{\rho}\nabla P + \boldsymbol{g} &= -\rho_0^{-1}(1+\alpha(T-T_0))(\nabla P_s + \nabla p) + \boldsymbol{g} \\ &= -\rho_0^{-1}\nabla P_s - \rho_0^{-1}\nabla P_s \alpha(T-T_0) - \rho_0^{-1}\nabla p - \rho_0^{-1}\nabla p\alpha(T-T_0) + \boldsymbol{g} \\ &\approx -\boldsymbol{g}\alpha(T-T_0) - \rho_0^{-1}\nabla p\end{aligned}$$

Where we neglected the second-order terms ($\sim \nabla pT$) and used that $\nabla P_s = \rho_0 \boldsymbol{g}$, cancelling out the gravity terms. With this, Eq. (13) becomes

$$\nu\nabla^2 \boldsymbol{u} = \boldsymbol{g}\alpha(T-T_0) + \rho_0^{-1}\nabla p \tag{15}$$

note that ∇p is only the non-hydrostatic pressure gradient.

We can take the curl of Eq. (15), using that $\nabla \times \boldsymbol{u} = \boldsymbol{\omega}$, that the curl of a gradient is always zero ($\nabla \times (\nabla f) = 0$), and that $\boldsymbol{g} = \nabla\phi$:

$$\begin{aligned}\nu\nabla^2\boldsymbol{\omega} &= \nabla \times (\boldsymbol{g}\alpha(T-T_0)) + \nabla \times (\rho_0^{-1}\nabla p) \\ &= \nabla \times [\boldsymbol{g}\alpha(T-T_0)] \\ &= \alpha(T-T_0)(\nabla \times \boldsymbol{g}) + \alpha\nabla T \times \boldsymbol{g} \\ &= \alpha\nabla T \times \boldsymbol{g}\end{aligned} \tag{16}$$

Now we assume that the flow in restricted to the x, z plane, meaning $\boldsymbol{u} = (u, 0, w) = \left(-\frac{\partial\psi}{\partial z}, 0, \frac{\partial\psi}{\partial x}\right)$ (where ψ is the stream function) and thus $\boldsymbol{\omega} = (0, \omega, 0) = (0, -\nabla^2\psi, 0)$. With $\boldsymbol{g} = (0, 0, g)$, Eq. 16 becomes (computing the cross product)

$$\nabla^2\omega = \frac{\alpha g}{\nu}\frac{\partial T}{\partial x} \tag{17}$$

From this equation, we can get an characteristic velocity for the convecting system.

$$\begin{aligned}\mathcal{O}(\nabla^2\omega) &= \mathcal{O}(\frac{\alpha g}{\nu}\frac{\partial T}{\partial x}) \\ \frac{U}{d^3} &= \frac{\alpha g}{\nu_0}\frac{T_1}{d} \\ U &= \frac{\alpha g T_1 d^2}{\nu_0}\end{aligned}$$

Where U, ν_0, and T_1 are characteristic values for velocity, kinematic viscosity, and temperature deviations from the adiabatic reference profile T_0, respectively.

For the scaling of pressure variations P_1 TO DO

2.3 The anelastic-liquid approximation

From now on, only the dimensionless variables will be used but with primes surpressed (i.e. all variables are assumed dimensionless). The dimensionless equation of state (from eq. 12) becomes

$$\rho = \rho_r[1 - \mu(T - T_s) + \mu(\rho_0 g d/K_T)P_1] \tag{18}$$

with $\mu = \alpha T_1$. If $\mu \to 0$, ρ reduces to its reference profile ρ_r (as is the case in the Boussinesq approximation).

Table 1: characteristic values of Jarvis and Mckenzie, 1980

Variable	Characteristic value	Dimensionless variable
depth	d	$(x', z') = \dfrac{(x, z)}{d}$
temperature	$T_1(= 1\mathrm{K})$	$T' = \dfrac{(T - T_0)}{T_1}$
density	ρ_0	$\rho'_r = \dfrac{\rho_r}{\rho_0}, \quad \rho = \dfrac{\rho}{\rho_0}$
velocity	$U = \dfrac{g\alpha T_1 d^2}{\nu_0}$	
kinematic viscosity	$\nu_0 = \eta/\rho_0$	
time	$\tau = d/U$	

3 Appendix 1: tables of symbols

Table 2: Symbols used by Jarvis and Mckenzie, 1980

d		depth of the convecting layer
H_T	$\frac{C_p}{g\alpha}$	temperature scale height of the liquid
C_p		specific heat at constant pressure
C_v		specific heat at constant volume
α		volume coefficient of thermal expansion
τ_{ij}		deviatoric stress tensor
$(\nabla T)_s$	$\frac{\alpha T}{\rho C_p}\nabla P$	adiabatic temperature gradient
K		thermal conductivity
H		rate of internal heat production
η	$\rho\nu$	dynamic viscosity
ν		kinematic viscosity
η_0	$\ll 1$	second or bulk viscosity
Γ	$\frac{\alpha K_s}{\rho C_p} = \frac{\alpha K_T}{\rho C_v}$	Grüneisen's parameter, assumed constant and $O(1)$
K_s		isentropic modulus of bulk compressibility
K_T		isothermal modulus of bulk compressibility
μ	αT_1	
α	$10^{-5} \quad {}^\circ K^{-1}$	volume coefficient of thermal expansion

Table 3: characteristic values of Jarvis and Mckenzie, 1980

depth	d
temperature	$T_1 = \frac{1}{\mathrm{K}} WHAT?? \rightarrow 1°\mathrm{K}$
density	ρ_0
velocity	$U = \frac{g\alpha T_1 d^2}{\nu_0}, \quad \nu_0 = \eta/\rho_0$help this is different from schubert
time	$\tau = d/U$

Table 4: Dimensionless variables in Jarvis and Mckenzie, 1980

(x', z')	$(x, z)/d$
ρ'_r	$\rho_r/\rho_0 = \exp[(1 - z')D/\Gamma], \quad D = d/H_T$
ρ'	ρ/ρ_0
T'	$(T - T_0)/T_1$
t'	t/τ
P'_1	$\dfrac{P_1 d}{U\eta} = P_1 \alpha T_1 \rho_0 g d, \quad P_1 = P - P_H$
κ'_0	$\kappa_0\nu_0/g\alpha T_1 d^3$, thermal diffusivity
ϵ'_0	$\epsilon_0\nu_0/g\alpha T_1^2 d$, energy generation rate, $\epsilon_0 = H/\rho_0 C_p$
λ	L/d, box width, L is the dimensional box width